

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026

Comprehensive Examination
(EC-3 Regular)

Course No. : CSI/SE/SSZG527
Course Title : Cloud Computing
Nature of Exam : Open Book
Weightage : 40%
Date : 24-05-2026 [AN]

No. of Pages	= 2
No. of Questions	= 8

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively.
3. Each answer should start from a fresh page.
4. Assumptions made if any, should be stated clearly at the beginning of your answer.
5. Avoid redundant terms/words /sentences. Save time to review every question and answer appropriately.

Q1	Suppose you are tasked with optimizing the migration of virtual machines (VMs) across different cloud regions. Each VM has a specific memory requirement and a priority level, while each cloud region has a limited amount of available memory and a cost per GB. Using your knowledge of optimization algorithms and cloud computing, design a function() (an algorithm clearly showing your approach) to migrate all VMs to the cloud regions with the lowest total cost while satisfying the memory constraints. Your function should take as input a list of VMs with their memory requirements and priority levels, and a list of cloud regions with their available memory and cost per GB. The output should be a dictionary/list mapping each VM to its assigned cloud region, along with the total cost of migration. For example, given the following input: VMs = [(100, 'high'), (50, 'medium'), (200, 'low')] Cloud Regions = [('Region1', 500, 0.1), ('Region2', 300, 0.2), ('Region3', 800, 0.05)] Your function should return a dictionary/list mapping that optimally assigns each VM to a cloud region, considering both the memory constraints and the cost. For the given input, the output will be <pre>{ 'assignments': { 'VM0': {'region': 'Region3', 'cost': 5.0}, 'VM1': {'region': 'Region3', 'cost': 2.5}, 'VM2': {'region': 'Region3', 'cost': 10.0} }, 'total_cost': 17.5 }</pre>	8
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	Show how your function would handle this example, and explain the reasoning behind your optimization approach. Be sure to consider the priority levels of the VMs and handle cases where it's not possible to migrate all VMs due to insufficient memory in the cloud regions. If it's not possible to migrate all VMs, your function should provide a clear indication of this, along with the optimally assigned VMs.	
Q2	SRL Corp hosts its analytics platform on AWS. Over six months, their data ingestion volume doubled. Which of the following is most likely to occur? Justify your choice (if no/wrong justification provided, no marks) a) AWS will automatically provide free storage upgrades. b) SRL Corp's costs will increase unless they optimize storage and compute usage. c) SRL Corp will continue paying the same amount regardless of usage. d) SRL Corp's system will automatically become more secure with higher data volume.	3
Q3	List any 3 factors that differentiate between non-relational database services and relational database services, by justifying each as to how can it be advantageous or a concern for cloud?	3
Q4	Comment on security of the cloud and security in the cloud with an appropriate example for each.	4
Q5	A company named Sattara is building an IoT system to host a set of visualization services over the web in a local network. It will require a webserver, PostgreSQL and RabbitMQ. They have decided to use Docker to deliver their services. They choose to do the following: a. Run an Ubuntu container and install webserver, PostgreSQL and RabbitMQ inside the container. b. Attach a public IP and DNS name to the container and release it to the public. Is this the right approach? If yes, why? If no, why and provide a better solution? You may assume other details required but you need to state it in your answer.	6
Q6	Suppose you are the chief architect of a cloud-based software as a service (SaaS) application. As a cloud architect, list the four levels of multi-tenancy discussed in the class. Now, identify any FOUR risks in multi-tenancy across these four levels and propose FOUR respective mitigation strategies.	6 (2 + 1*4 = 4)
Q7	You are given the following Dockerfile: Identify any TWO problems in this Dockerfile and propose a respective best practice. <pre>FROM ubuntu:latest RUN apt-get update RUN apt-get install -y python3 RUN apt-get install -y curl COPY . /app</pre>	5 (2 + 3)

	RUN pip install -r /app/requirements.txt	
	CMD ["python3", "/app/app.py"]	
Q8	A distributed system has 7 nodes. Find the minimum quorum size required to maintain consistency with appropriate justification. Now, if a network partition leaves 3 nodes isolated and 4 nodes connected, explain whether the system can continue processing writes?	5 (2 + 3)